

JATIN SATYAM

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EDUCATION

M.S. Robotics & Autonomous Systems (Systems Eng.), Arizona State University · GPA 3.67/4.0 Expected May 2027
B.E. Mechanical Engineering with Major (Honors) in Electric Vehicles, University of Mumbai 2020 – 2024

EXPERIENCE

Research Assistant Aug 2026 – Present
Robotics Systems Instructional Lab, Arizona State University

- Built the ROS 2 Jazzy / Gazebo Harmonic model of a two-robot semiconductor wafer-handling cell (Dobot M1 Pro SCARA, myCobot Pro 600, conveyor) running the full wafer pick-transfer-place cycle autonomously; raised real-time factor 0.971 to 0.984, repaired both vendor URDFs, and shipped the live pipeline as a ROS 2 package verified in Docker on Humble / Gazebo Fortress
- Brought up a read-only real-to-sim digital shadow mirroring both physical arms live over the lab LAN, in Gazebo (tracking within 1.9 degrees / 7.6 mm M1 Pro, 4.4 degrees Pro 600) and in NVIDIA Isaac Sim 6.1 (bench-proven, worst error 0.5 / 2.2 degrees over a 461 s recorded cycle); a read-only Modbus TCP client rebuilds the PLC's six-state sequence against a Micro850 simulator

Mechatronics Engineer Jan 2025 – Jul 2025
Paras Defence & Space Technologies Ltd., Navi Mumbai, India

- Designed and iterated Gifford-McMahon cryocooler mechanisms in SolidWorks and NX, refining moving-assembly tolerances and interfaces for reliable cold-head displacer motion and manufacturability
- Led performance testing of 3 cryogenic cooling assemblies across 25+ instrumented runs, characterizing thermal losses and translating findings into mechanism-level design changes that cut prototype rework 12%

CFD Engineer Aug 2024 – Nov 2024
MARS Exploration Pvt. Ltd, Remote

- Performed 3D CFD analysis of NACA 23012 and NACA 4412 airfoils in ANSYS Fluent 2024 R2 (10-degree AoA, 48 m/s) - geometry, boundary-layer-refined meshing with quality checks, solver setup, and convergence monitoring

R&D Engineer May 2023 – Aug 2023
TriQual Cryogenics Pvt Ltd, Navi Mumbai, India

- Contributed to mechanical design and vibration control for Stirling-cycle cryocoolers operating near 70 K

PROJECTS

Space Robotics & AI ↗ · ROS 2 Jazzy · Gazebo Harmonic · PX4 SITL 2026

- Aerial-ground autonomy on real Mars terrain: a UAV maps HiRISE Jezero Crater terrain into a Bayesian occupancy grid (0.25 m/cell) that a 6-wheel rover navigates with A* replanning; terrain-relative navigation at 2 Hz over K=294 pose candidates
- Also: Drone search, mapping & precision landing (PX4, ArUco, RTAB-Map SLAM; 150/150), Apollo 17 photogrammetry & 3D Gaussian splatting, Cart-pole LQR, Boustrophedon coverage controller

Robot Arm Manipulation Lab ↗ · Python · OpenCV · Dobot Magician Lite 2025

- Natural-language manipulation with vision & LLM control on a physical Dobot Magician Lite: Llama 3.3 70B on Groq parses commands, HSV vision locates objects, validated motion primitives execute; camera-to-robot calibration to +/-2 mm
- Also: YOLO detection & palletizing, Vision tic-tac-toe with Minimax, Grid-maze navigation with BFS, FK/IK three ways, Suction cup vs gripper pick & place

Vision-Guided Aerial Autonomy ↗ · Simulink · Flight Control · Machine Vision 2026

- Controlled landing on a moving platform: Simulink controllers deployed to a physical quadrotor track a platform sliding along a track from above and land on it in motion, in one continuous take (position held within ~5 cm of the platform in simulation)
- Also: Area scan & landing on a visual target, Line following with end-of-line landing, Square trajectory tracking, two laps, Attitude control & stable hover, six Simulink iterations

SKILLS

Programming: Python (NumPy, SciPy, OpenCV, Open3D), C++, C, MATLAB, Bash, SQL
Software: ROS 2 (Humble/Jazzy), Gazebo, NVIDIA Isaac Sim, PX4 SITL, MuJoCo, Simulink, RTAB-Map, Docker, SolidWorks, Siemens NX, ANSYS, Git & GitHub
Other Skills: Kinematics, Motion Planning, A* Planning, SLAM, Occupancy Mapping, PID, LQR, State-Space Modeling

AWARDS & RECOGNITION

- 2nd Prize, National Level Project Poster Competition, IEI-FCRIT 2024 - gyro motor rotor (CALIBRE 2k24)
- Winner, MLH Best Use of Auth0 - Meridian, Innovation Hacks 2.0, ASU, Apr 2026
- Co-inventor, invention disclosure filed with ASU's Skysong Innovations - SipSafe Smart Cup, 2026